

Validity of the restricted (primal) and relaxed (dual) bounds – DD C++
DD: results_cpp_30_07_2026_09-55-57.csv | exact: results_exact_cpp_23_05_2026_14-38-42.csv | Gurobi: results_gurobi_cpp_30_07_2026_09-56-16.csv
Reference optimum: exact DD (not reduced) when available, otherwise Gurobi with gap 0 [SOC Knapsack: exact DD only]
Rule (max): Restricted <= optimum <= Relaxed. (min): Relaxed <= optimum <= Restricted. Relative tolerance 1e-06.
17589 checkable configurations over 490 instances with a known optimum (405 exact DD, 85 Gurobi gap 0). Each cell: bounds that hold / bounds checked. Violations: 13.
Python and C++ agree on the diagrams (python_cpp_agreement_table.py), so only C++ is checked.

By DD type - bounds that hold / bounds checked

	Restricted	RelaxPriority	RelaxGrouping	Total
Knapsack	348/348	346/346	345/345	1039/1039
Independent Set	492/492	484/484	490/490	1466/1466
Set Cover	3888/3888	3888/3888	3888/3888	11664/11664
SOC Knapsack	1080/1080	1080/1080	1080/1080	3240/3240
Sequencing	60/60	56/60	51/60	167/180
Total	5868/5868	5854/5858	5854/5863	17576/17589

By max width - bounds that hold / bounds checked

	500	1000	2000	5000	10000	20000	Total
Knapsack	180/180	180/180	180/180	177/177	168/168	154/154	1039/1039
Independent Set	246/246	246/246	246/246	246/246	244/244	238/238	1466/1466
Set Cover	1944/1944	1944/1944	1944/1944	1944/1944	1944/1944	1944/1944	11664/11664
SOC Knapsack	540/540	540/540	540/540	540/540	540/540	540/540	3240/3240
Sequencing	30/30	30/30	29/30	27/30	25/30	26/30	167/180
Total	2940/2940	2940/2940	2939/2940	2934/2937	2921/2926	2902/2906	17576/17589

Bound violations – 13 configurations, sorted by relative violation within each problem

sequencing_atsp_br17_n16_seed2.txt	[Sequencing]	RelaxGrouping w=5000 reduced=true sort=true	-> value 5,218 vs optimum 4,495 (Exact DD)	RelaxGrouping (min) above the optimum by 723 (16.0810%)
sequencing_atsp_br17_n16_seed2.txt	[Sequencing]	RelaxGrouping w=2000 reduced=true sort=true	-> value 5,041 vs optimum 4,495 (Exact DD)	RelaxGrouping (min) above the optimum by 546 (12.1441%)
sequencing_atsp_br17_n16_seed3.txt	[Sequencing]	RelaxGrouping w=10000 reduced=true sort=true	-> value 4,526 vs optimum 4,050 (Exact DD)	RelaxGrouping (min) above the optimum by 476 (11.7502%)
sequencing_atsp_br17_n16_seed2.txt	[Sequencing]	RelaxGrouping w=10000 reduced=true sort=true	-> value 4,921 vs optimum 4,495 (Exact DD)	RelaxGrouping (min) above the optimum by 426 (9.4751%)
sequencing_atsp_br17_n16_seed2.txt	[Sequencing]	RelaxGrouping w=20000 reduced=true sort=true	-> value 4,873 vs optimum 4,495 (Exact DD)	RelaxGrouping (min) above the optimum by 378 (8.4075%)
sequencing_atsp_br17_n16_seed5.txt	[Sequencing]	RelaxGrouping w=20000 reduced=true sort=true	-> value 5,144 vs optimum 4,805 (Exact DD)	RelaxGrouping (min) above the optimum by 339 (7.0537%)
sequencing_atsp_br17_n16_seed3.txt	[Sequencing]	RelaxGrouping w=10000 reduced=false sort=true	-> value 4,310 vs optimum 4,050 (Exact DD)	RelaxGrouping (min) above the optimum by 260 (6.4182%)
sequencing_atsp_br17_n16_seed2.txt	[Sequencing]	RelaxGrouping w=20000 reduced=false sort=true	-> value 4,714 vs optimum 4,495 (Exact DD)	RelaxGrouping (min) above the optimum by 219 (4.8710%)
sequencing_atsp_br17_n16_seed5.txt	[Sequencing]	RelaxGrouping w=10000 reduced=true sort=true	-> value 5,008 vs optimum 4,805 (Exact DD)	RelaxGrouping (min) above the optimum by 203 (4.2239%)
sequencing_atsp_br17_n16_seed2.txt	[Sequencing]	RelaxPriority w=5000 reduced=true sort=true	-> value 4,641 vs optimum 4,495 (Exact DD)	RelaxPriority (min) above the optimum by 146 (3.2473%)
sequencing_atsp_br17_n16_seed2.txt	[Sequencing]	RelaxPriority w=20000 reduced=true sort=true	-> value 4,608 vs optimum 4,495 (Exact DD)	RelaxPriority (min) above the optimum by 113 (2.5133%)
sequencing_atsp_br17_n16_seed2.txt	[Sequencing]	RelaxPriority w=10000 reduced=true sort=true	-> value 4,550 vs optimum 4,495 (Exact DD)	RelaxPriority (min) above the optimum by 55 (1.2233%)
sequencing_atsp_br17_n16_seed2.txt	[Sequencing]	RelaxPriority w=5000 reduced=false sort=true	-> value 4,538 vs optimum 4,495 (Exact DD)	RelaxPriority (min) above the optimum by 43 (0.9564%)